

FAIR data & ALF

Streamlining ALF/pyALF for data management and integrating with NOMAD

Jefferson Stafusa E. Portela

11.11.2022

University of Würzburg

Key collaborators: Jonas Schwab (Uni. Würzburg), Jose M. Pizarro (FAIRmat)

Data management: process to model, gather, store, clean, integrate, and access data

FAIR data: data that remain open and usable

NOMAD: repository and data laboratory for materials science

ALF: general, open-source code for auxiliary-field quantum Monte Carlo simulations

- Motivation
- FAIR principles
- NOMAD
- ALF
- NOMAD & ALF
- Outlook

Old data is valuable:

Old data is valuable:

- In HEP:
 - JADE data (DESY, –1986): 2004 Nobel Prize (Wilczek, Politzer) for asymptotic freedom
 - Tevatron data (Fermilab, –2011): reanalysis published in 2022 Science paper with 265 citations

Old data is valuable:

- In HEP:
 - JADE data (DESY, –1986): 2004 Nobel Prize (Wilczek, Politzer) for asymptotic freedom
 - Tevatron data (Fermilab, –2011): reanalysis published in 2022 Science paper with 265 citations
- For anything that's not static:
 - Astronomy (ephemeris, variable stars, exoplanets search)
 - Biodiversity (extinction risk evaluation, hidden dependencies – upward shift of alpine plants in Switzerland, vegetation dependency on sand dunes in northern Germany on wild ungulates)
 - Climate (sea-level measurements, ice cover)
 - Earth Sciences (landslide and flood hazards, long-term effects of interventions – effect of reforestation with non-native trees on Cape Town's reservoirs)
 - ...

Benefits of adequate, open data management:

- enable FAIR data (i.e., data that remain open and usable)

Benefits of adequate, open data management:

- enable FAIR data (i.e., data that remain open and usable)
- increase replicability

Benefits of adequate, open data management:

- enable FAIR data (i.e., data that remain open and usable)
- increase replicability
- avoid redundancies, duplicated efforts

Benefits of adequate, open data management:

- enable FAIR data (i.e., data that remain open and usable)
- increase replicability
- avoid redundancies, duplicated efforts
- support new discoveries

Benefits of adequate, open data management:

- enable FAIR data (i.e., data that remain open and usable)
- increase replicability
- avoid redundancies, duplicated efforts
- support new discoveries
- facilitate collaboration

Benefits of adequate, open data management:

- enable FAIR data (i.e., data that remain open and usable)
- increase replicability
- avoid redundancies, duplicated efforts
- support new discoveries
- facilitate collaboration
- fulfill funders' and legal requirements (required by all US federal agencies, European Union and UK councils, etc.)

Benefits of adequate, open data management:

- enable FAIR data (i.e., data that remain open and usable)
- increase replicability
- avoid redundancies, duplicated efforts
- support new discoveries
- facilitate collaboration
- fulfill funders' and legal requirements (required by all US federal agencies, European Union and UK councils, etc.)
- elicit more citations (+10%—50%)

Def.: FAIR data: data that remain usable.

Def.: FAIR data: data that remain usable.

To be **FAIR**, data must satisfy:

Def.: FAIR data: data that remain usable.

To be **FAIR**, data must satisfy:

- Findable**
- unique ID
 - rich metadata

Def.: FAIR data: data that remain usable.

To be **FAIR**, data must satisfy:

Findable

- unique ID
- rich metadata

Accessible

- permanent retrievability
- open, free, standard protocol

Def.: FAIR data: data that remain usable.

To be **FAIR**, data must satisfy:

- Findable**
- unique ID
 - rich metadata

- Accessible**
- permanent retrievability
 - open, free, standard protocol

- Interoperable**
- formal, accessible language
 - qualified references

Def.: FAIR data: data that remain usable.

To be **FAIR**, data must satisfy:

- | | |
|----------------------|--|
| Findable | <ul style="list-style-type: none">• unique ID• rich metadata |
| Accessible | <ul style="list-style-type: none">• permanent retrievability• open, free, standard protocol |
| Interoperable | <ul style="list-style-type: none">• formal, accessible language• qualified references |
| Reusable | <ul style="list-style-type: none">• open, rich metadata• community standards |

Def.: FAIR data: data that remain usable.

To be **FAIR**, data must satisfy:

- | | |
|----------------------|--|
| Findable | <ul style="list-style-type: none">• unique ID• rich metadata |
| Accessible | <ul style="list-style-type: none">• permanent retrievability• open, free, standard protocol |
| Interoperable | <ul style="list-style-type: none">• formal, accessible language• qualified references |
| Reusable | <ul style="list-style-type: none">• open, rich metadata• community standards |

Based on: Wilkinson et al, Sci Data **3**, 160018 (2016), DOI: [10.1038/sdata.2016.18](https://doi.org/10.1038/sdata.2016.18).

NOMAD Laboratory

enables FAIR sharing and use of materials science data

Mar 23, 2021 [April 8, 2021: CECAM Session on Data Driven Science with an Overview Talk by Claudia Draxl](#)

Publish

- Publish your data with our without embargo, get a DOI, and share data with others.
- We support input and output files of most electronic-structure codes.
- Watch our [video tutorial](#) on how to upload and publish data.

UPLOAD

Explore

- Search for [materials](#) (Encyclopedia) or [calculations](#) (Repository).
- All raw and processed data can be downloaded and used under the [CC BY 4.0](#).
- Watch our [video tutorials](#) on how to use the Encyclopedia and Repository.

MATERIALS

CALCULATIONS

Analyze

- Analyze data with [Jupyter notebooks](#) directly on NOMAD servers (Artificial Intelligence (AI) Toolkit).
- Access all data programmatically via [NOMAD API](#) or [OPTIMADE API](#).
- Watch our [video tutorial](#) on how to use the NOMAD API.

AI TOOLKIT TUTORIALS

Algorithms Lattice Fermions

Algorithms for Lattice Fermions: general open-source code for auxiliary-field quantum Monte Carlo simulations

General Hamiltonian

- single-body operators,
- squares of single-body operators
- single-body operators coupled to a scalar field

Flexibility

- model
- Bravais lattice
- observables (equal-time and time-displaced)

High Performance

- parallelization (MPI, OpenMP)
- near-optimal single-core performance
- restart facilities
- HDF5 data files

Ease of Use

- Python interface: pyALF
- extensive documentation
- Predefined Structures
- scripts and templates

Advanced Features

- parallel tempering
- ground state projective QMC
- global Monte Carlo updates
- continuous fields
- Langevin dynamics updates
- stochastic Maximum Entropy
- symmetric Trotter decomposition
- Dimer-Dimer correlations
- Cotunneling (Kondo)
- Rényi Entropy

*Algorithms
Lattice
Fermions*



Algorithms for Lattice Fermions: general, open-source code for auxiliary-field quantum Monte Carlo simulations
keywords: condensed matter, solid state physics

NOMAD: creates, collects, stores, and cleanses computational materials science data, computed by the most important materials-science codes available today; 100+ million calculations

*Algorithms
Lattice
Fermions*

Algorithms for Lattice Fermions: general, open-source code for auxiliary-field quantum Monte Carlo simulations
keywords: condensed matter, solid state physics



NOMAD: creates, collects, stores, and cleanses computational materials science data, computed by the most important materials-science codes available today; 100+ million calculations

Win-win situation:

Adapting ALF's output to increase machine readability, and extending NOMAD to process ALF data improves both programs.

ALF data in NOMAD current status: Beta version (schema and parser still work in progress)



QMC test

upload id: zLRnpX70QKeU9-_YFtJhfQ



1 Prepare and upload your files

To prepare your data, simply use **zip** or **tar** to create a single file that contains all your files as they are. These .zip/.tar files can contain subdirectories and additional files. NOMAD will search through all files and identify the relevant files automatically. Each uploaded file can be **up to 32GB** in size, you can have **up to 10 unpublished uploads** simultaneously. Your uploaded data is not published right away. Find more details about uploading data in our [documentation](#) or visit our [FAQs](#). The following codes are supported: **Atomistic codes:** [ABACUS](#), [ABINIT](#), [ALF](#), [AMS](#), [ASAP](#), [Amber](#), [BigDFT](#), [CASTEP](#), [CHARMM](#), [CP2K](#), [CPMD](#), [CRYSTAL](#), [DFTB+](#), [DL_POLY](#), [DMol3](#), [Elk](#), [FHI-aims](#), [FLEUR](#), [FPLO](#), [GAMESS](#), [GPAW](#), [GROMACS](#), [GROMOS](#), [GULP](#), [Gaussian](#), [LAMMPS](#), [MOPAC](#), [Molcas](#), [NAMD](#), [NWChem](#), [ONETEP](#), [ORCA](#), [Octopus](#), [OpenMX](#), [Psi4](#), [Qball](#), [Qbox](#), [QuantumATK](#), [QuantumESPRESSO](#), [SIESTA](#), [TURBOMOLE](#), [Tinker](#), [VASP](#), [WIEN2k](#), [Wannier90](#), [YAMBO](#), [exciting](#), [libAtoms](#), **Workflow managers:** [AFLOW](#), [ASR](#), [Atomate](#), [ElaStic](#), [FHI-vibes](#), [LOBSTER](#), [MOFStructures](#), [QuantumEspressoPhonon](#), [QuantumEspressoEPW](#), [phonopy](#), **Database managers:** [EELSDB](#), [NeXus](#), [OpenKIM](#). Click the code to get more specific information about how to prepare your files.



CLICK OR DROP FILES

▼ /

▼ 1

confout_0.h5

data.h5 mainfile [alf](#) ...

info

parameters

Metadata

method name

QMC

workflow name

single_point

program name

ALF

model name

Square Hubbard

type

ALF

name

X16 ALF simulation

comment

no comment

references

authors

Sheldon Cooper

datasets

no datasets

mainfile

data.h5/1



entry id

hasJMWU0-ODbFeshFEzpE90aEKg



Material

Composition

formula

X16

structural type

quasi-2D

material name

unavailable

elements

X

number of elements

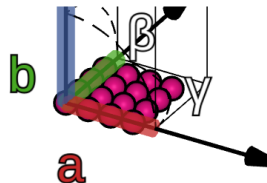
1 (unary)

Symmetry

no data

Original ▾

Structure



Lattice parameters

a	b	c
1 Å	1 Å	4 Å
α	β	γ
90 °	90 °	90 °
cell volume		
4 Å ³		

[VIEW IN ENCYCLOPEDIA](#)

Run

section

SUB SECTIONS

program

method

system

▼ calculation

0

1

Calculation

section

SUB SECTIONS

x_alf_den_eq

x_alf_den_tau

x_alf_ener_scal

x_alf_green_eq

x_alf_green_tau

x_alf_kin_scal

x_alf_part_scal

x_alf_pot_scal

x_alf_spint_eq

x_alf_spinxy_eq

x_alf_spinxy_tau

x_alf_spinz_eq

x_alf_spinz_tau

QUANTITIES

system_ref = reference ...

method_ref = reference ...

▼ calculations_ref

0

x_alf_observable_eqtau_data

section

QUANTITIES

x_alf_obser = $213 \times 1 \times 1 \times 1 \times 16 \times 2$ tensor

x_alf_back = $213 \times 1 \times 2$ tensor

x_alf_sign = 213 vector

x_alf_obser

quantity

VALUE

1	1.84805	1
2.00×10^{-12}	-0.01230	2.00×10^{-12}

PREV 0 NEXT

PREV 0 NEXT

PREV 0 NEXT

PREV 0 NEXT

(NBins × Norbs × Norbs × Ntau × Nlatt × 2)

run

sub section definition

DESCRIPTION

Every section run represents a single call of a program.

BASE SECTION

ArchiveSection

SUB SECTION DEFINITIONS

program

time_run

message

method repeats

system repeats

calculation repeats

QUANTITY DEFINITIONS

calculation_file_uri

clean_end

raw_id

starting_run_ref

<>

method

sub section definition

DESCRIPTION

Section containing the various parameters that define the theory and the approximations (convergence, thresholds, etc.) behind the calculation.

SUB SECTION DEFINITIONS

dft

projection

gw

qmc

tb

force_field

k_mesh

electronic

scf

atom_parameters repeats

molecule_parameters repeats

basis_set repeats

<>

qmc

sub section definition

DESCRIPTION

Section containing the various parameters that define a QMC calculation.

QUANTITY DEFINITIONS

model_name

u_0

u_nn

chemical_potential

inverse_temperature

dtau

n_wraps

n_bins

n_sweeps

<>

u_0

quantity definition

DESCRIPTION

Hubbard interaction in units of the nearest neighbor hopping.

PROPERTIES

type: float64 (numpy)

shape: []

<>

- **This year:**
 - Beta version of ALF parser available at our local instance of NOMAD
 - ALF output partially adapted to NOMAD
- **Next year:**
 - ALF parser available at the central NOMAD
 - ALF output fully streamlined
 - improved parsers
 - more sophisticated analysis of ALF data available at NOMAD

- **This year:**
 - Beta version of ALF parser available at our local instance of NOMAD
 - ALF output partially adapted to NOMAD
- **Next year:**
 - ALF parser available at the central NOMAD
 - ALF output fully streamlined
 - improved parsers
 - more sophisticated analysis of ALF data available at NOMAD

Thanks!